Neural-Network-Based Time-Scale Modification Algorithm on Audio Signal
基於神經網路的時間拉伸演算法

Progress Report 2021.4.16

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Time-scale Modification

Time-scale Modification

- TSM
  - Handcrafted
    - Time domain
    - Spectrogram
  - Data-driven
    - Parallel data
    - Unparalleled data

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No current work on this

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Our project would focus on this

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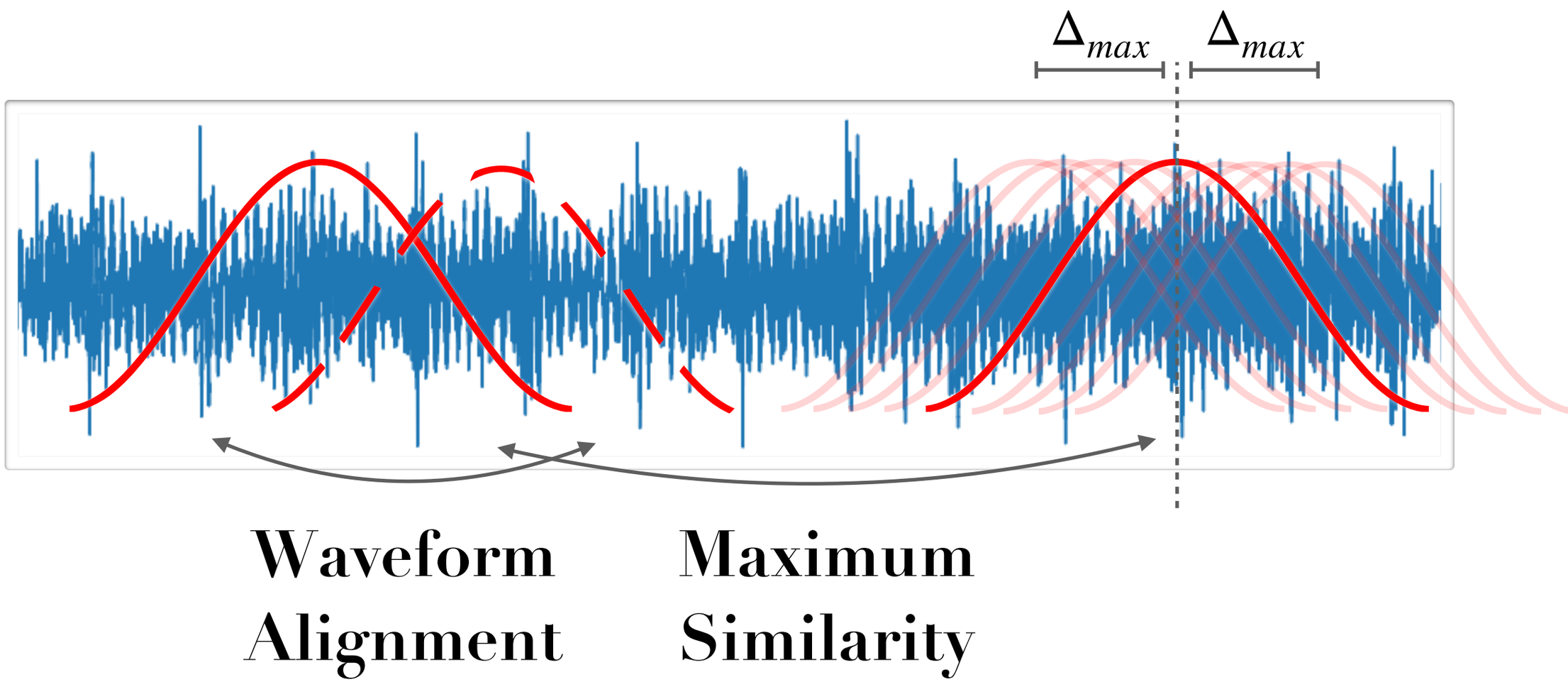
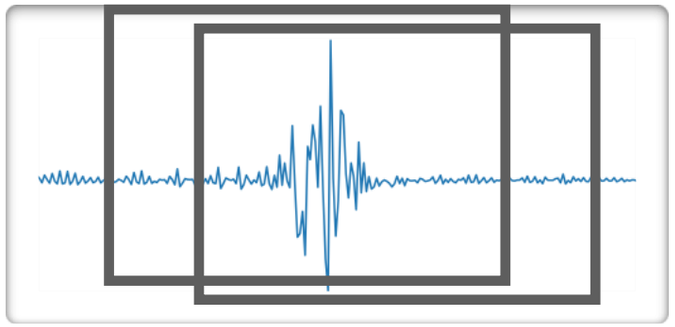
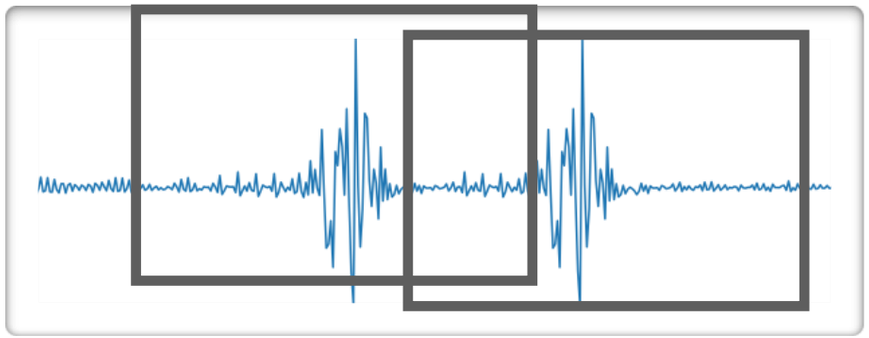
WSOLA


Δ_{max}
Δ_{max}
Waveform Alignment
Maximum Similarity

WSOLA

- Pros:
  - More performant on percussions
  - Faster than SOLAFS
- Cons:
  - Transient doubling/skipping
  - Can only preserve the most prominent harmonic frequency





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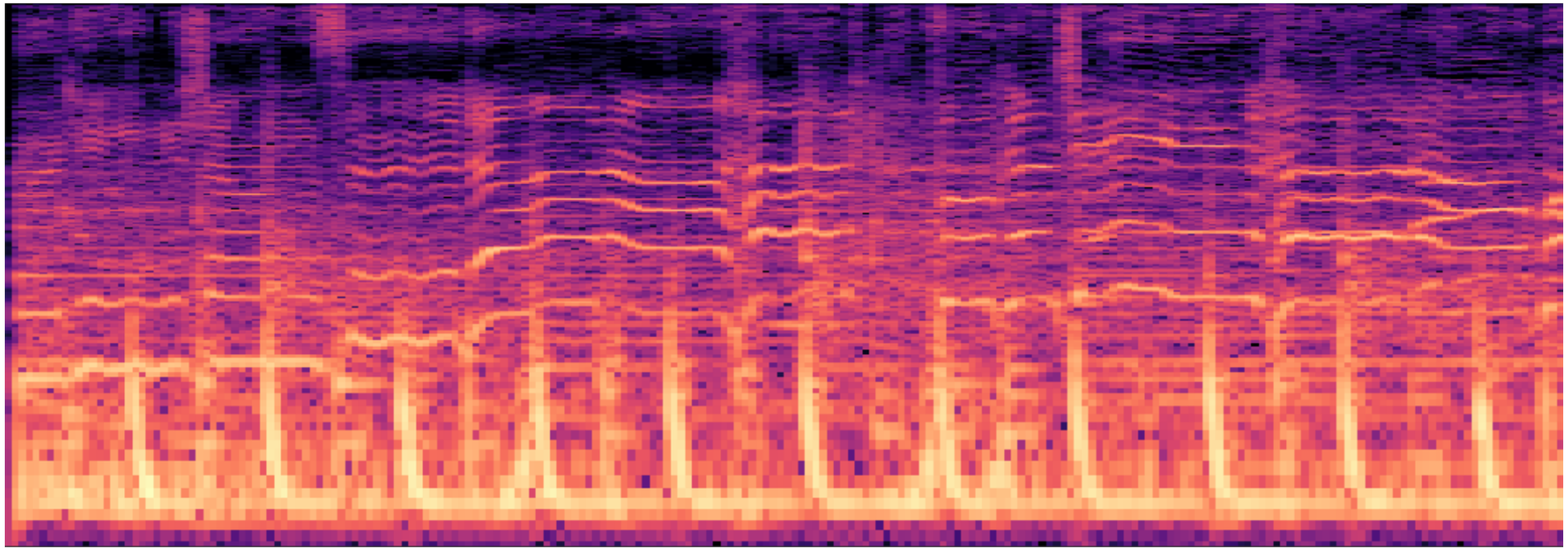
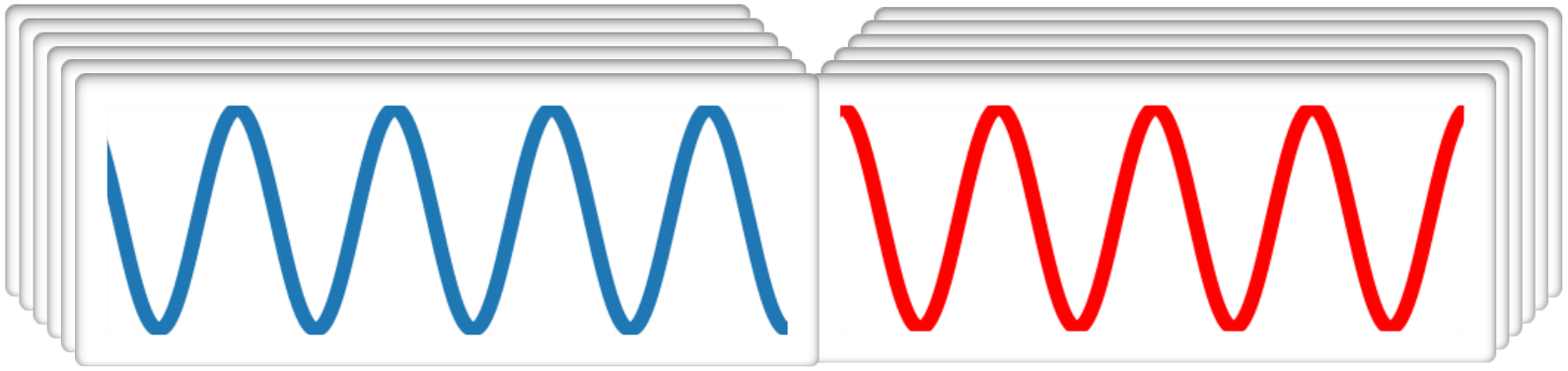
PV-TSM

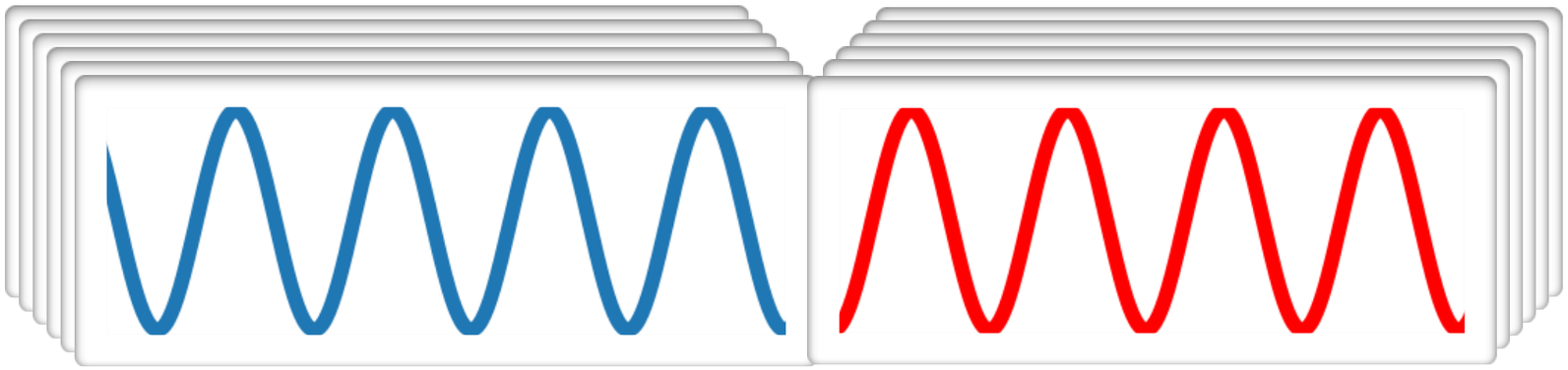
- TSM based on phase vocoder
- Spectrogram



PV-TSM



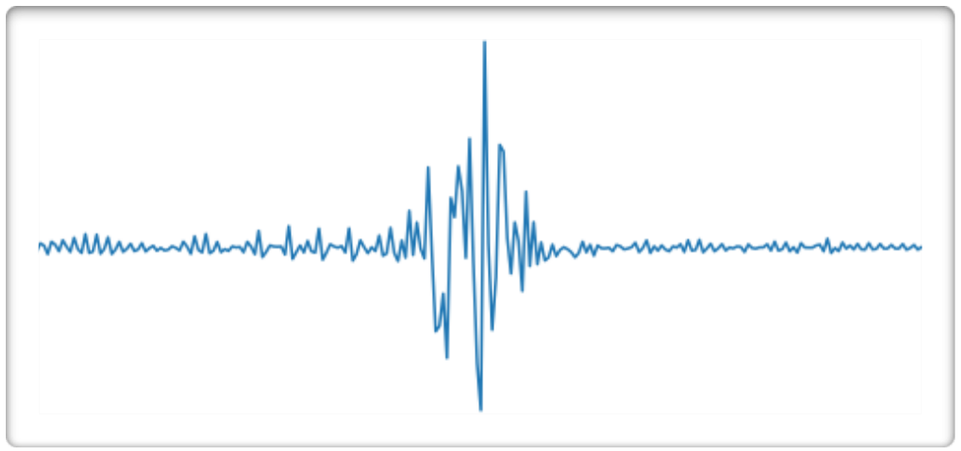
Phase discontinuous

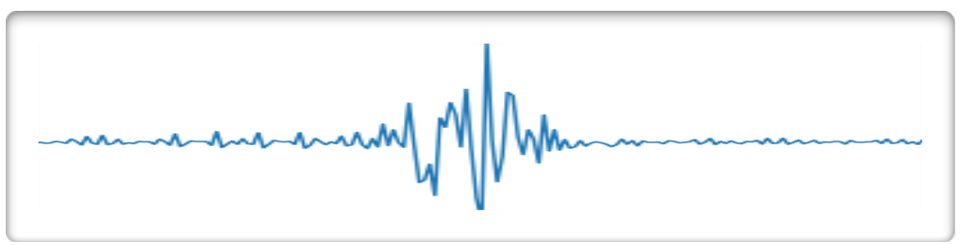


Phase alignment

PV-TSM

- Pros:
  - More performant on harmonics
  - Preserve multiple frequencies
- Cons:
  - Smeared transient
  - Lose vertical phase coherence





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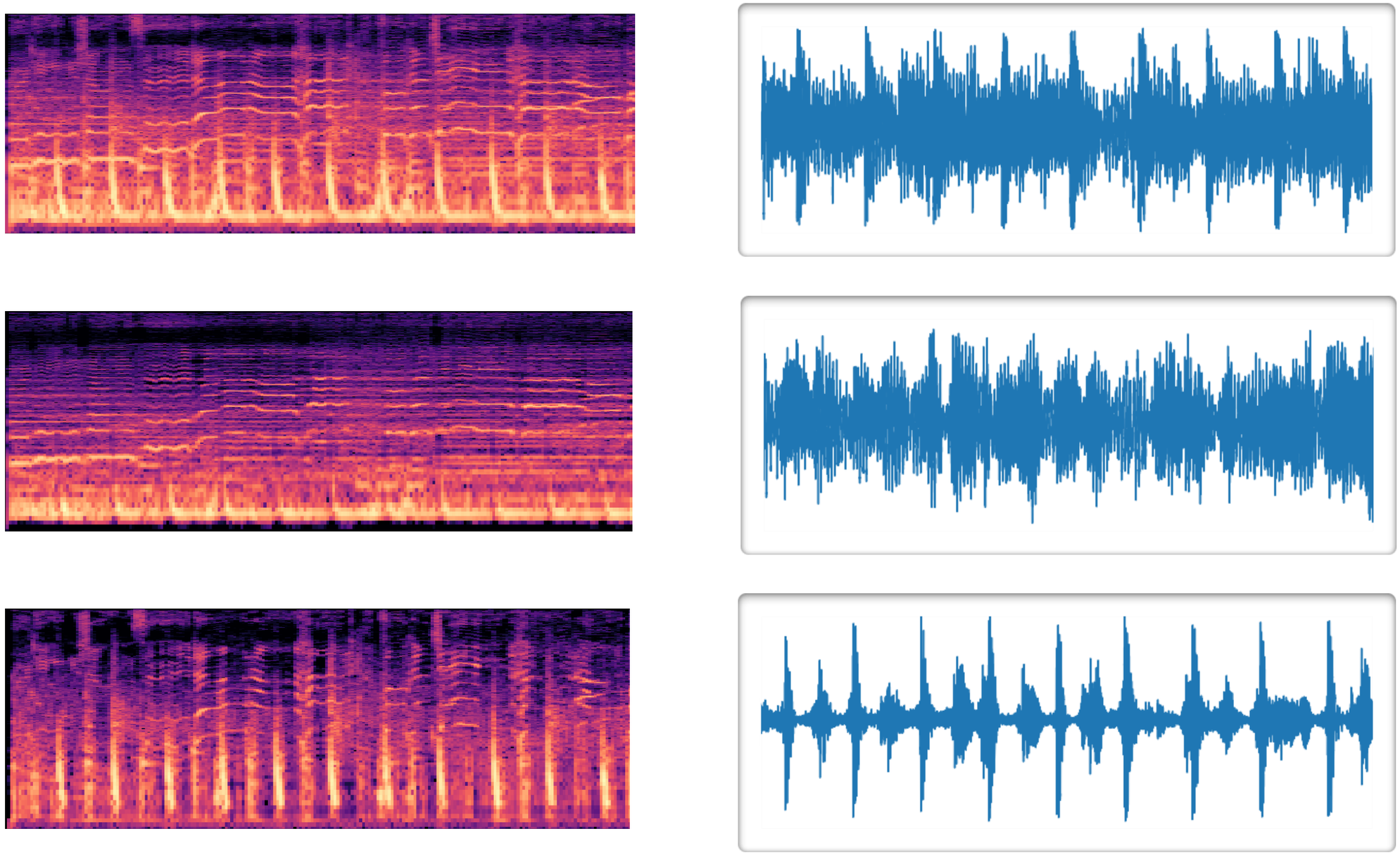
HPSS



Liu, 2018, Denoising Auto-encoder with Recurrent Skip Connections and Residual Regression for Music Source Separation.

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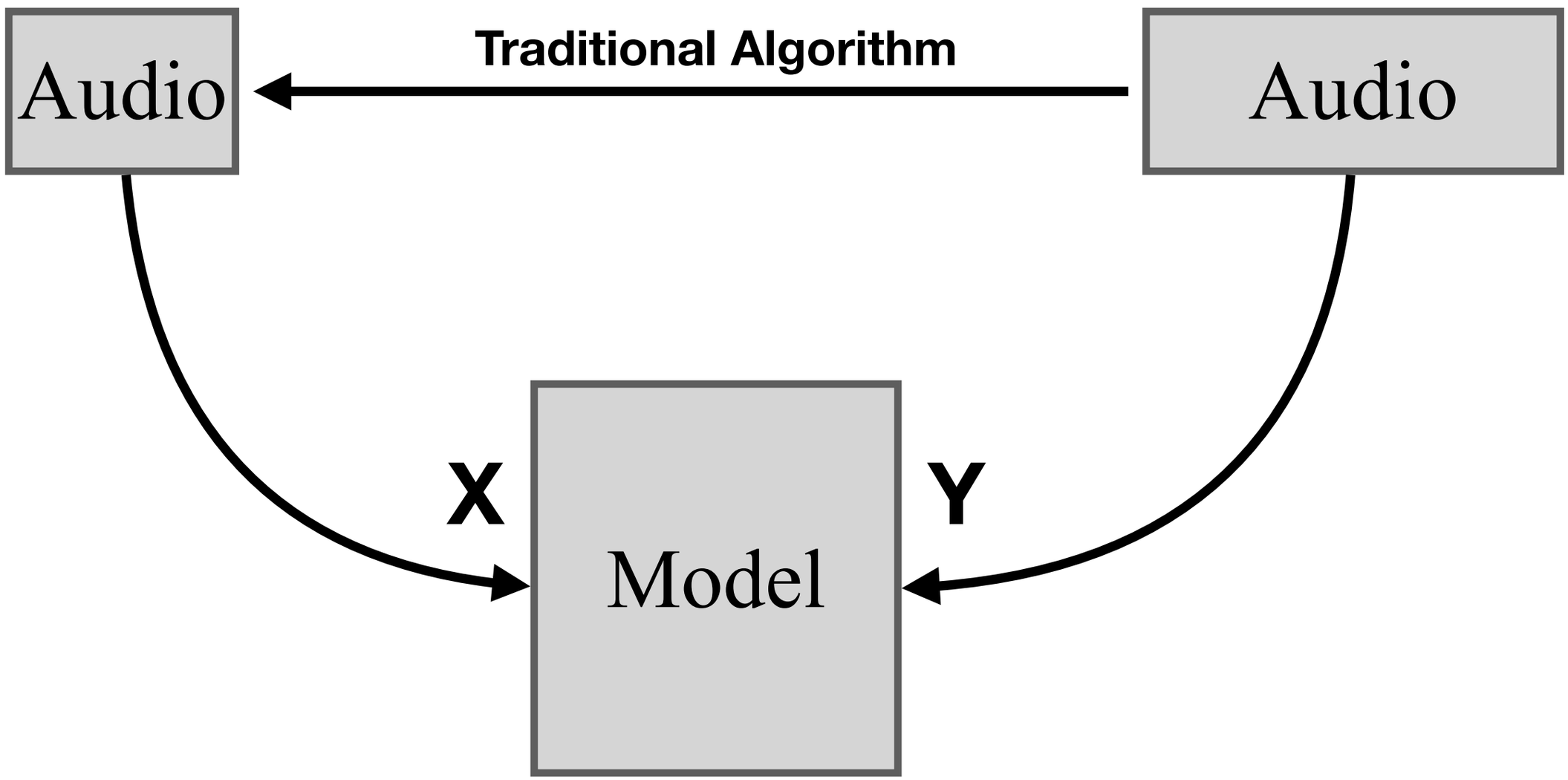
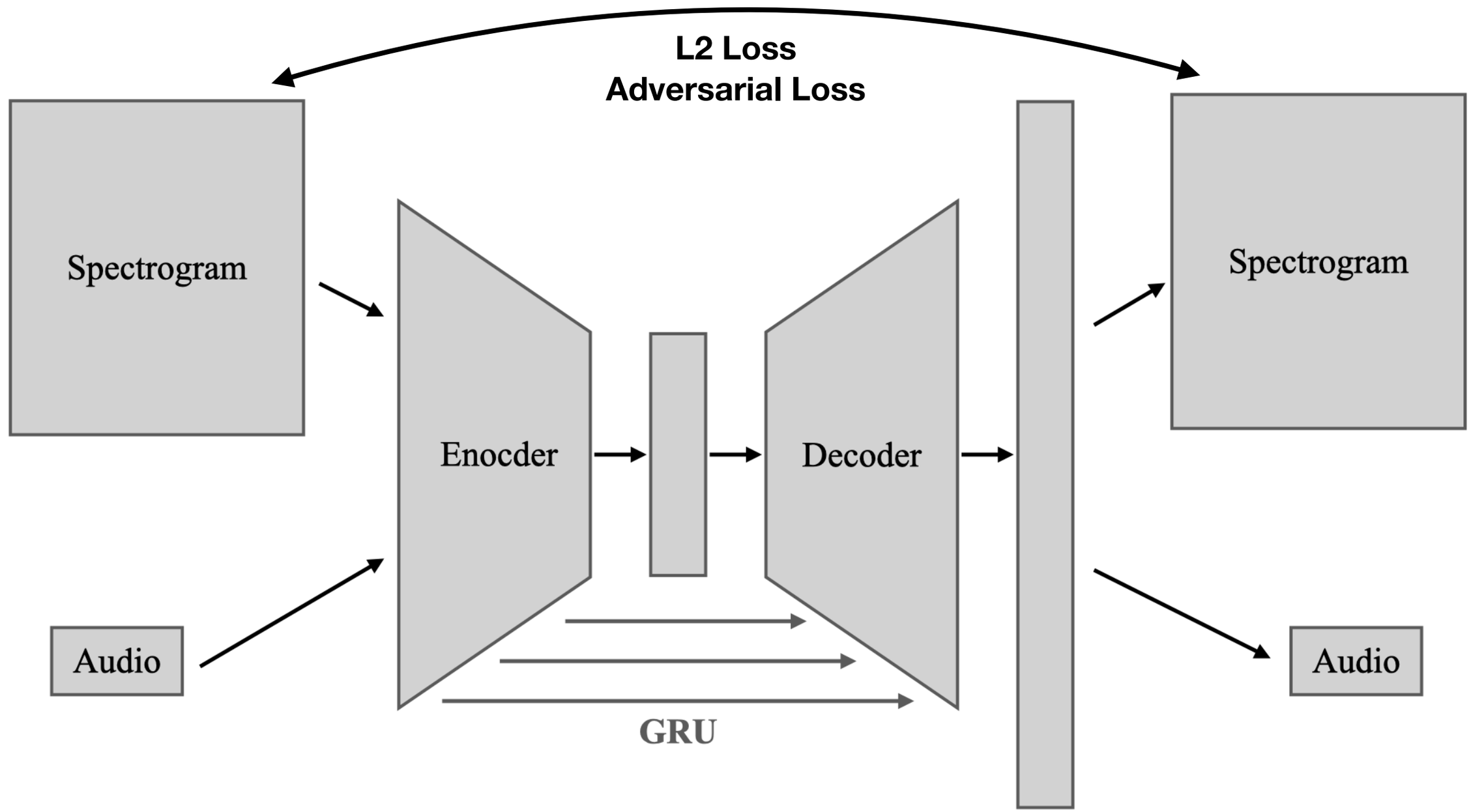
Parallel data

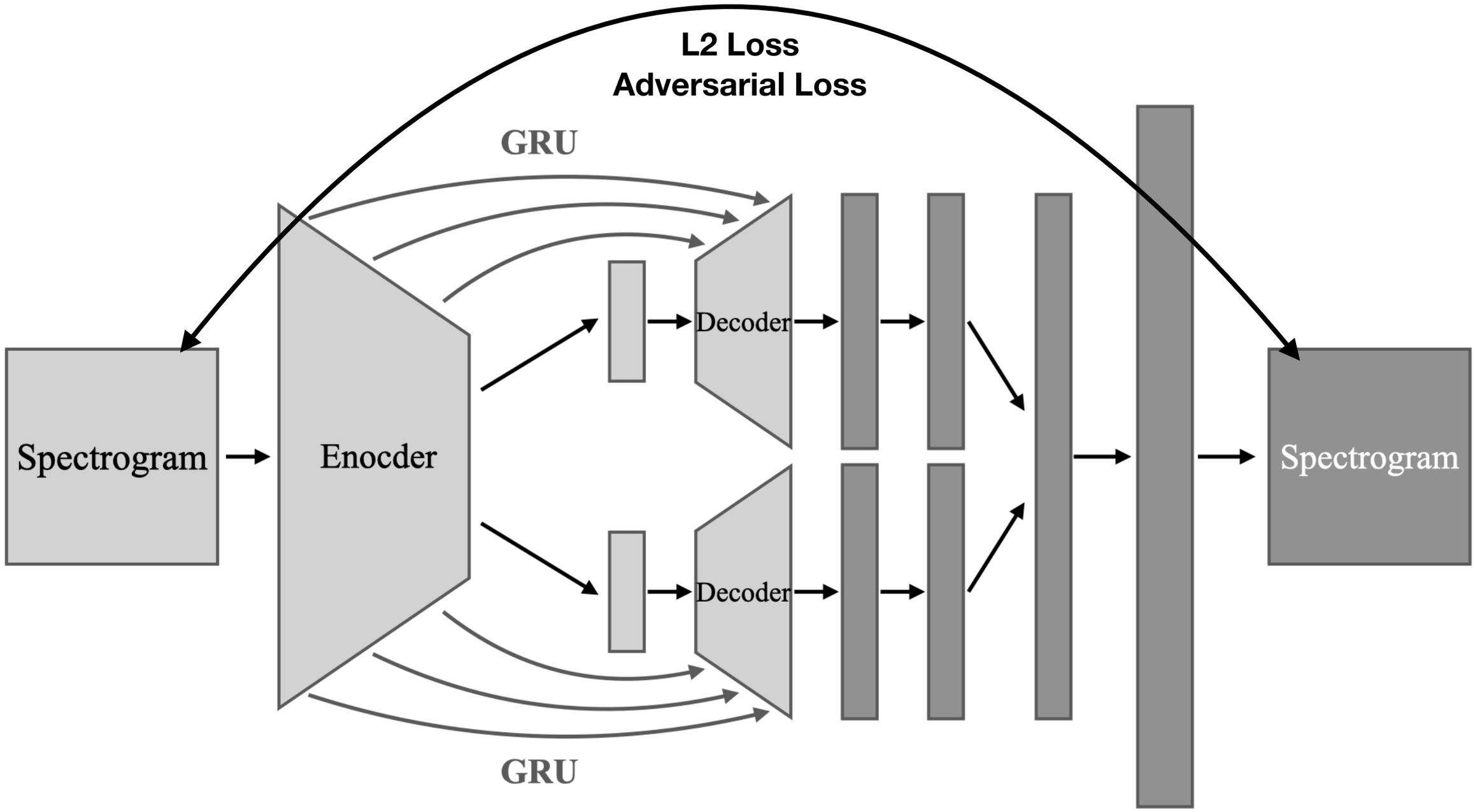
Parallel data



One-staged


L2 Loss
Adversarial Loss
Spectrogram
Enocder
Decoder
Spectrogram
Audio
Audio
GRU

Two-staged


L2 Loss
Adversarial Loss
GRU
Spectrogram
Enocder
Decoder
Decoder
Spectrogram
GRU

Parallel Data

- Pros:
  - More supervised, more robust
- Cons:
  - Fixed rate (and can only slow down)
  - Dataset produced by the traditional TSM may not reflect the real data distribution

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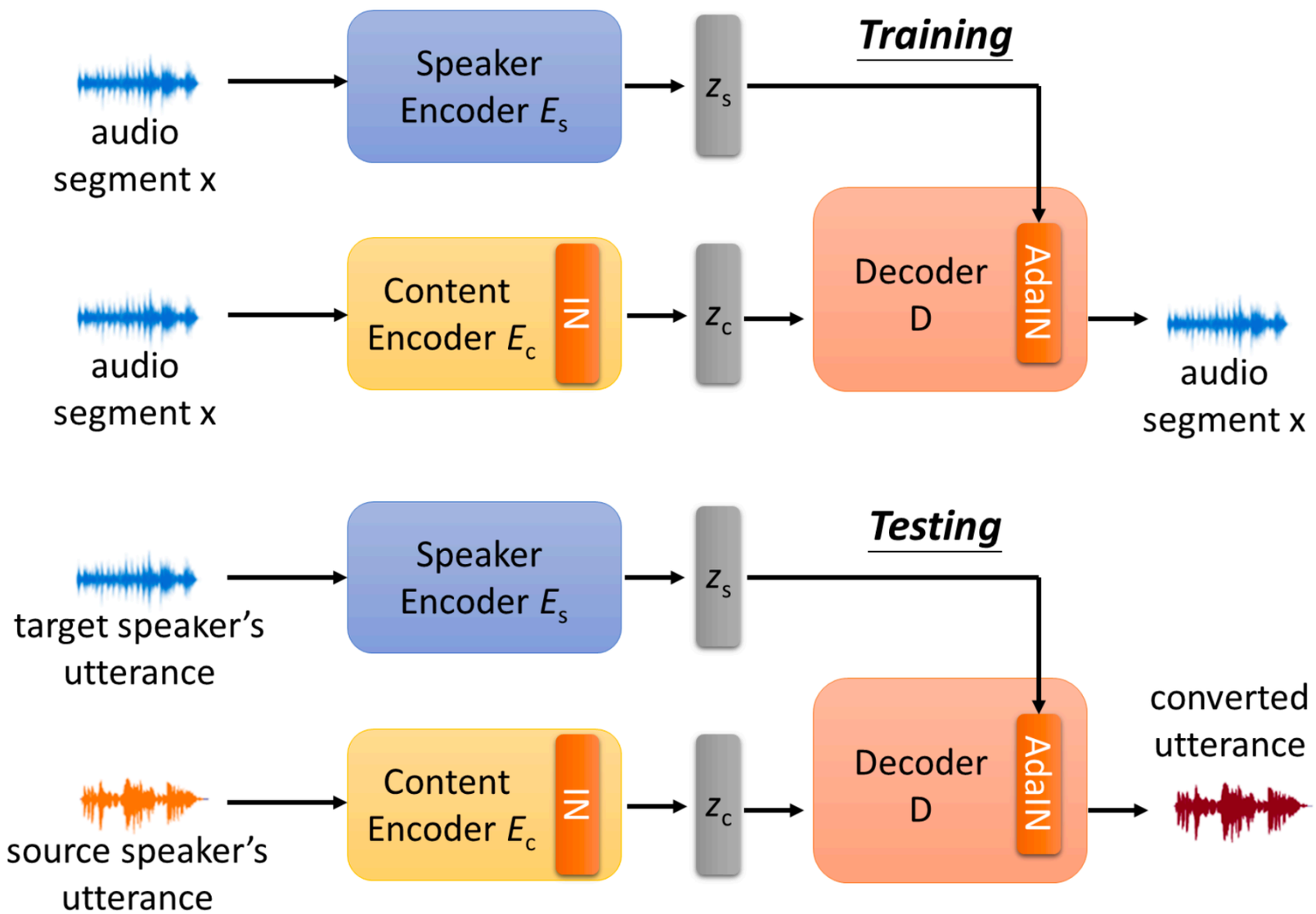
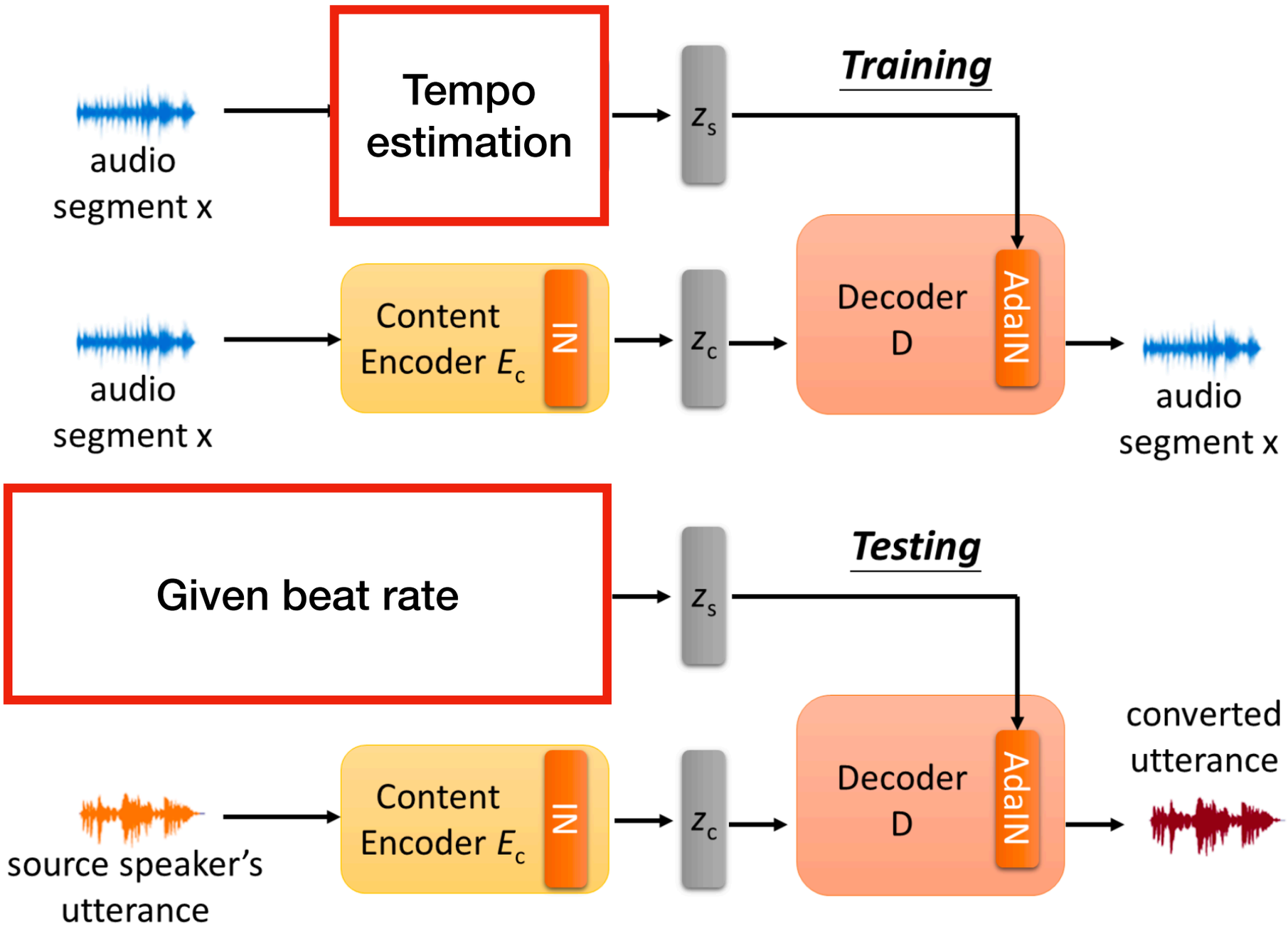
Feature Disentangle

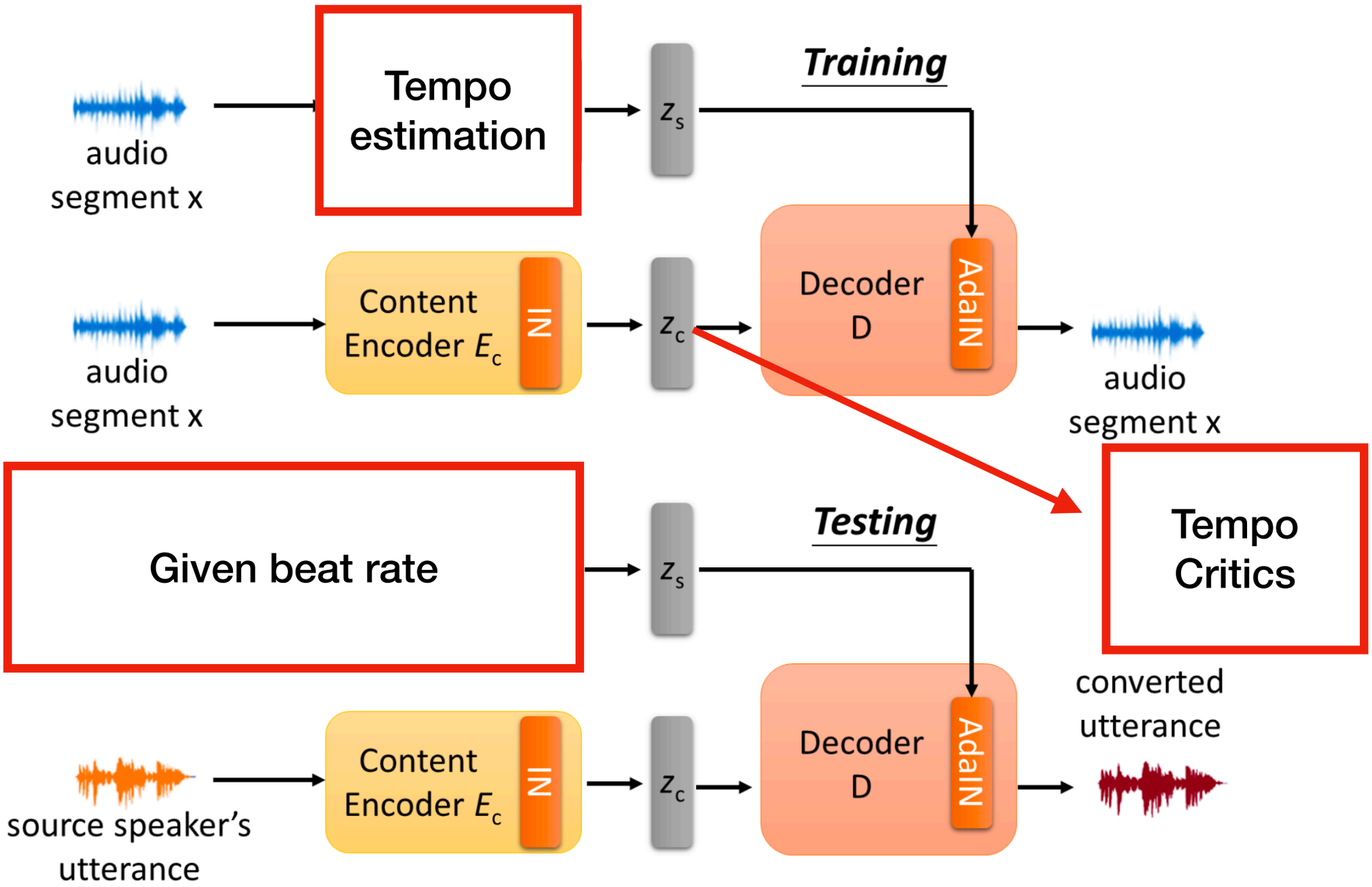


Chou, 2019, One-shot Voice Conversion by Separating Speaker and Content Representations with Instance Normalization

Feature Disentangle



Feature Disentangle



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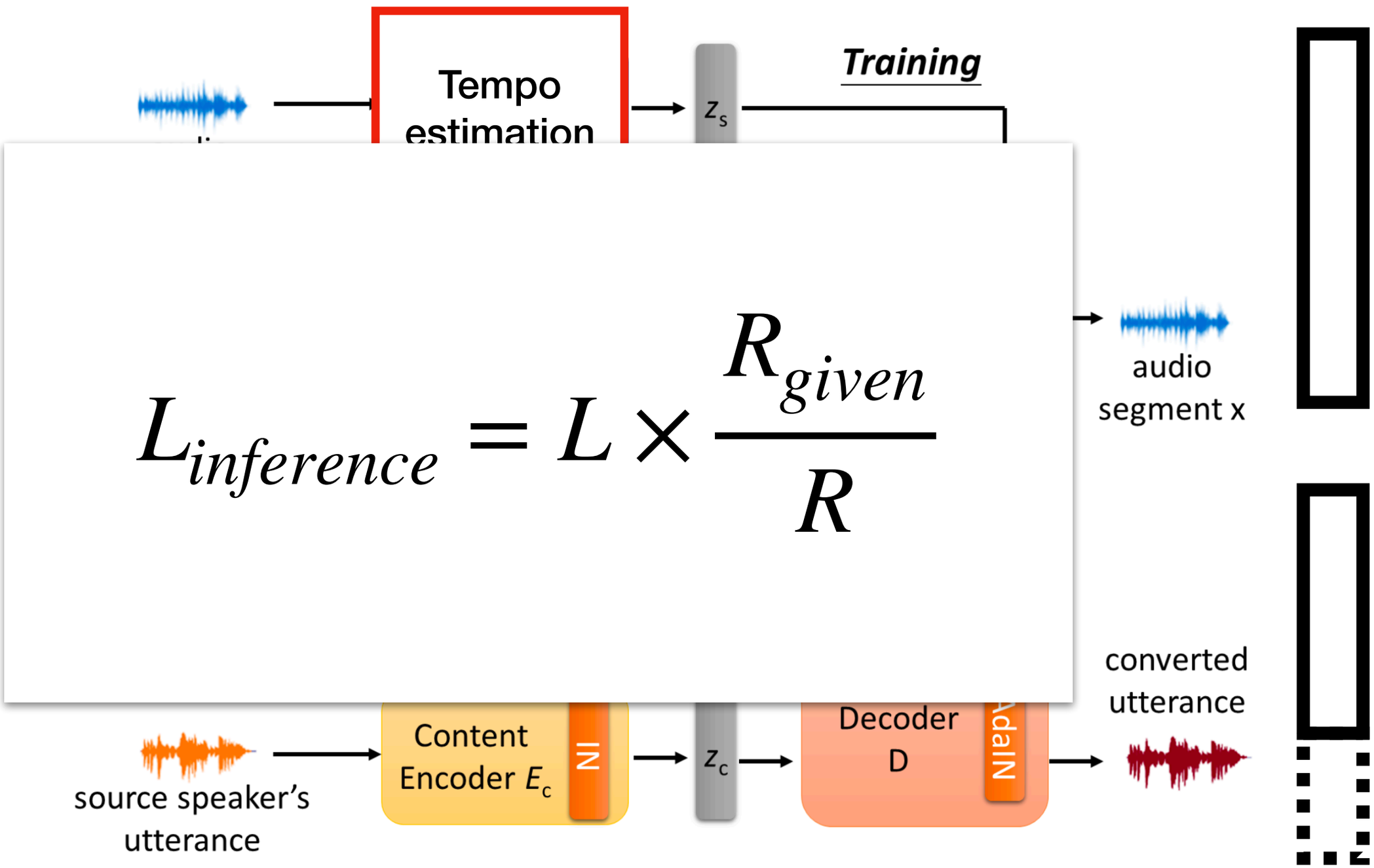
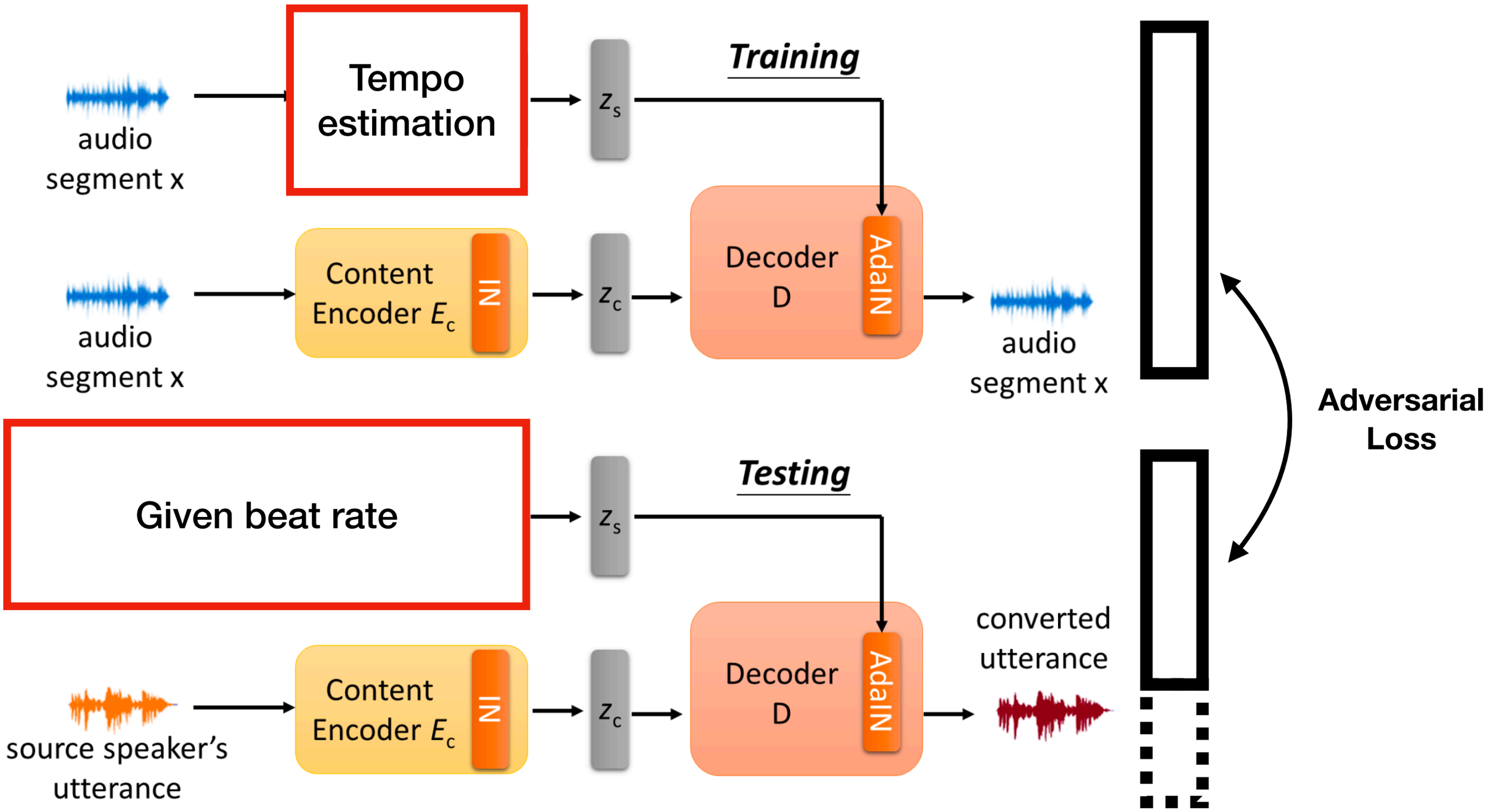


$$L_{inference} = L \times \frac{R_{given}}{R}$$

Feature Disentangle



Feature Disentangle

- Pros:
  - Variable rate
  - Dataset is easier to produce
- Cons:
  - Content encoder may still contains tempo information

Thanks for Listening